

Lecture 4: Union Bound, Maximum of Sub-Gaussians, Bootstrap, and Synthesis

The $\sqrt{\log p}$ Cost of Multiplicity,
and a Unified Decision Framework

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1 Recap and Roadmap

The first three lectures built the two pillars of this course:

- **Lectures 1–2 (Asymptotic):** LLN, CLT, MLE consistency, asymptotic normality ($\sqrt{n}$ -rate), Delta method, and three failure modes (misspecification, heavy tails, high dimensions).
- **Lecture 3 (Non-asymptotic):** Cramér–Chernoff method, Hoeffding’s inequality, Bernstein’s inequality, and the sub-Gaussian/sub-exponential families.

Today we use both pillars in two concrete, self-contained applications, and then step back to synthesize the two paradigms.

Lecture 4 roadmap:

1. **Union bound and simultaneous inference:** controlling the maximum over many random variables; the $\sqrt{\log p}$ price of multiplicity.
2. **Maximum of sub-Gaussian variables:** the key result $\mathbb{E}[\max_i X_i] \leq \sigma\sqrt{2\log p}$ and its comparison with the asymptotic order statistic.
3. **Bootstrap: asymptotic meets finite-sample:** the bootstrap as a middle ground between the two paradigms.
4. **Synthesis: a decision framework:** when to use which paradigm, open problems, and the big picture.

The Unifying Question of Lecture 4

Both paradigms ultimately ask: *how much data is enough?*

- Asymptotic: enough for the approximation error to be negligible.
- Non-asymptotic: enough for an explicit probability guarantee to hold.

These are genuinely different questions with genuinely different answers. Today we learn to choose between them — and to recognize when they agree.

2 Union Bound and Simultaneous Inference

2.1 The union bound

The union bound (Boole’s inequality) is the simplest tool for controlling the probability that *any one* of many bad events occurs.

Theorem 2.1 — Union Bound (Boole’s Inequality)

For any (possibly dependent) events $A_1, \dots, A_m$,

$$\mathbb{P}\left(\bigcup_{j=1}^m A_j\right) \leq \sum_{j=1}^m \mathbb{P}(A_j).$$

Proof. By induction and the sub-additivity of probability measures: $\mathbb{P}(A_1 \cup \dots \cup A_m) \leq \mathbb{P}(A_1) + \mathbb{P}(A_2 \cup \dots \cup A_m) \leq \dots \leq \sum_{j=1}^m \mathbb{P}(A_j)$. $\square$

Remark — The Union Bound Requires No Independence

The union bound holds for *arbitrarily dependent* events. This generality is its strength: we pay no price for dependence, only for the number of events m .

2.2 Application: simultaneous confidence intervals

Suppose we construct m confidence intervals, each with individual coverage $1 - \alpha$. What is the probability that *all* of them are simultaneously correct?

Example 2.2 — Multiple Testing Problem

Let $X_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_j, \sigma^2)$ for $j = 1, \dots, p$, each with n observations. Consider the p individual 95% CIs for μ_j : $\hat{\mu}_j \pm 1.96 \sigma / \sqrt{n}$.

Asymptotic guarantee (per interval): $\mathbb{P}(\mu_j \notin \text{CI}_j) = 0.05$ for each fixed j .

Simultaneous guarantee (union bound):

$$\mathbb{P}(\exists j : \mu_j \notin \text{CI}_j) \leq \sum_{j=1}^p \mathbb{P}(\mu_j \notin \text{CI}_j) = 0.05 p.$$

For $p = 100$, this exceeds 1 — the bound is vacuous. To achieve simultaneous coverage $1 - \delta$, we need each individual level to be $\alpha = \delta/p$ (**Bonferroni correction**).

2.3 Bonferroni correction and its non-asymptotic interpretation

Theorem 2.3 — Bonferroni Simultaneous CI

Let $X_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_j, \sigma^2)$ for $j = 1, \dots, p$, each with n_j observations. For each j , define $\text{CI}_j = \hat{\mu}_j \pm z_{\delta/(2p)} \sigma / \sqrt{n_j}$. Then

$$\mathbb{P}(\forall j \in \{1, \dots, p\} : \mu_j \in \text{CI}_j) \geq 1 - \delta.$$

The half-width of each interval is $z_{\delta/(2p)} \sigma / \sqrt{n_j}$.

Proof. By the union bound, $\mathbb{P}(\exists j : \mu_j \notin \text{CI}_j) \leq \sum_{j=1}^p \mathbb{P}(\mu_j \notin \text{CI}_j) = \sum_{j=1}^p \delta/p = \delta$. $\square$

Remark — The Cost of Multiplicity: $\sqrt{\log p}$ Factor

For large p , the Bonferroni quantile $z_{\delta/(2p)}$ grows at rate $\sqrt{\log p}$ (up to constant factors):

$$z_{\delta/(2p)} = \Phi^{-1}(1 - \delta/(2p)) \approx \sqrt{2 \log(p/\delta)}.$$

This is the $\sqrt{\log p}$ **price of multiplicity**: controlling p tests simultaneously costs a factor of $\sqrt{\log p}$ in CI width compared to a single test. We will see the same $\sqrt{\log p}$ factor appear as a mathematical necessity in Section 3 — it is not an artifact of the union bound.

Remark — Bonferroni is Non-Asymptotic

Theorem 2.3 uses the exact Gaussian distribution of $\hat{\mu}_j$, so the simultaneous coverage $\geq 1 - \delta$ holds for *every* $n_j \geq 1$. When the X_{ij} are non-Gaussian, replace the per-test CI by a Hoeffding-based CI (Lecture 3), and the union bound argument is unchanged. The result: an exactly valid simultaneous bound for all n , at the cost of the $\sqrt{\log p}$ factor in width.

3 Maximum of Sub-Gaussian Variables

3.1 The problem: bounding the maximum

Let $X_1, \dots, X_p$ be random variables (possibly dependent). How large is $\max_{1 \leq j \leq p} X_j$?

This question arises naturally in:

- **High-dimensional statistics:** the maximum of p test statistics;
- **Model selection:** the best-fitting model among p candidates;
- **Simulation:** the largest of p random draws.

3.2 Upper bound via union bound

Theorem 3.1 — Maximum of Sub-Gaussian Variables

Let $X_1, \dots, X_p$ be (possibly dependent) random variables, each sub-Gaussian with parameter σ^2 and $\mathbb{E}[X_j] = 0$. Then for all $u \geq 0$:

$$\mathbb{P}\left(\max_{1 \leq j \leq p} X_j \geq u\right) \leq p e^{-u^2/(2\sigma^2)}.$$

In particular, setting $u = \sigma \sqrt{2 \log(p/\delta)}$:

$$\mathbb{P}\left(\max_{1 \leq j \leq p} X_j \geq \sigma \sqrt{2 \log(p/\delta)}\right) \leq \delta.$$

Proof. By the union bound and the sub-Gaussian tail bound (Lecture 3, Proposition 4.4(ii)):

$$\mathbb{P}\left(\max_j X_j \geq u\right) = \mathbb{P}\left(\bigcup_{j=1}^p \{X_j \geq u\}\right) \leq \sum_{j=1}^p \mathbb{P}(X_j \geq u) \leq p e^{-u^2/(2\sigma^2)}.$$

Setting $p e^{-u^2/(2\sigma^2)} = \delta$ and solving for u gives the stated threshold. $\square$

3.3 Lower bound: the result is tight

The upper bound $\sigma\sqrt{2\log p}$ is not just a technicality — it reflects the true order of magnitude of the maximum.

Theorem 3.2 — Expected Maximum of i.i.d. Sub-Gaussians

Let $X_1, \dots, X_p \stackrel{\text{iid}}{\sim}$ with $\mathbb{E}[X_j] = 0$ and $X_j \sim \text{subG}(\sigma^2)$. Then:

$$\mathbb{E}\left[\max_{1 \leq j \leq p} X_j\right] \leq \sigma\sqrt{2\log p}.$$

Moreover, if $X_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$, then as $p \rightarrow \infty$,

$$\frac{\mathbb{E}[\max_j X_j]}{\sigma\sqrt{2\log p}} \rightarrow 1,$$

so the bound is **asymptotically tight** for Gaussians.

Proof of the upper bound. Use the *Laplace transform trick*: for any $t > 0$,

$$\mathbb{E}\left[\max_j X_j\right] = \frac{1}{t} \mathbb{E}[\log e^{t \max_j X_j}] \leq \frac{1}{t} \log \mathbb{E}[e^{t \max_j X_j}]$$

by Jensen's inequality applied to $\log$ (concave). Since $e^{t \max_j X_j} = \max_j e^{t X_j} \leq \sum_j e^{t X_j}$:

$$\mathbb{E}\left[\max_j X_j\right] \leq \frac{1}{t} \log \left(\sum_{j=1}^p \mathbb{E}[e^{t X_j}] \right) \leq \frac{1}{t} \log(p e^{\sigma^2 t^2/2}) = \frac{\log p}{t} + \frac{\sigma^2 t}{2}.$$

Optimizing over $t > 0$: minimize $\log p/t + \sigma^2 t/2$ by setting $t^* = \sqrt{2\log p}/\sigma$, which gives

$$\mathbb{E}\left[\max_j X_j\right] \leq \sigma\sqrt{2\log p}. \quad \square$$

Remark — The Proof Strategy

The proof uses three ideas from this course in sequence:

1. **Jensen's inequality** to convert $\mathbb{E}[\max]$ to $\log \mathbb{E}[\exp]$.
2. **Union bound** ($\max_j e^{tX_j} \leq \sum_j e^{tX_j}$) to split over j .
3. **Sub-Gaussian MGF bound** ($\mathbb{E}[e^{tX_j}] \leq e^{\sigma^2 t^2/2}$) from Lecture 3.
4. **Optimization over t** — the same Cramér–Chernoff trick.

This is a prototype for many proofs in high-dimensional probability.

3.4 Asymptotic perspective on the maximum

The asymptotic paradigm gives a complementary, sharper picture when the X_j are i.i.d. Gaussian.

Theorem 3.3 — Asymptotic Distribution of the Maximum (Gumbel)

Let $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Define the normalizing constants

$$a_p = \sqrt{2 \log p}, \quad b_p = \sqrt{2 \log p} - \frac{\log \log p + \log(4\pi)}{2\sqrt{2 \log p}}.$$

Then as $p \rightarrow \infty$,

$$a_p \left(\max_{1 \leq j \leq p} X_j - b_p \right) \xrightarrow{d} \text{Gumbel}(0, 1),$$

where the Gumbel CDF is $F(x) = \exp(-e^{-x})$.

Remark — Non-Asymptotic vs. Asymptotic for the Maximum

- **Non-asymptotic bound (Theorem 3.1):** valid for all p and all sub-Gaussian distributions. Gives the correct order $\sigma\sqrt{2 \log p}$ with a simple proof. Holds for *dependent* X_j too. Does not give the exact distribution or sharp constants.
- **Asymptotic result (Theorem 3.3):** valid only as $p \rightarrow \infty$ and only for i.i.d. Gaussians (or distributions in their max-domain of attraction). Gives the exact limiting distribution, enabling precise p -values and sharper critical values. Requires independence.
- Both confirm the same key message: $\max_j X_j$ is of order $\sqrt{\log p}$ — bounded above and below by constant multiples of it. The non-asymptotic bound is the right tool for guarantees; the asymptotic result is the right tool for inference.

Example 3.4 — Comparing the Two Approaches Numerically

Let $X_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, $p = 1000$, $\delta = 0.05$.

Non-asymptotic bound (Theorem 3.1): $u = \sqrt{2 \log(1000/0.05)} = \sqrt{2 \log 20000} \approx \sqrt{19.8} \approx 4.45$.

$$\mathbb{P}\left(\max_j X_j \geq 4.45\right) \leq 0.05. \quad (\text{guaranteed})$$

Asymptotic (Gumbel) critical value: The 95th percentile of $\text{Gumbel}(b_p, 1/a_p)$ for $p = 1000$ gives a threshold ≈ 3.72 .

$$\mathbb{P}\left(\max_j X_j \geq 3.72\right) \approx 0.05. \quad (\text{approximate})$$

Exact simulation: The true 95th percentile of $\max_j X_j$ for $p = 1000$ i.i.d. $\mathcal{N}(0, 1)$ is approximately 3.72.

The non-asymptotic bound (4.45) is conservative by a factor of ≈ 1.2 , but it required no distributional assumption beyond sub-Gaussianity and works for all p . The Gumbel approximation (3.72) is more accurate but requires $p \rightarrow \infty$, i.i.d. Gaussian, and gives only approximate coverage.

4 Bootstrap: Where the Two Paradigms Meet

4.1 Motivation

The asymptotic paradigm requires knowing (or estimating) the asymptotic variance. For complex statistics — ratios, quantiles, functionals of the empirical distribution — this variance may be analytically intractable. The **bootstrap** approximates the sampling distribution of a statistic using the empirical distribution $\hat{P}_n$ as a surrogate for P .

Definition 4.1 — The Bootstrap

Given data $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P$, the **nonparametric bootstrap** proceeds:

1. Draw B bootstrap samples $X_1^{*(b)}, \dots, X_n^{*(b)} \stackrel{\text{iid}}{\sim} \hat{P}_n$ (i.e., sample with replacement from $\{X_1, \dots, X_n\}$).
2. Compute the statistic $T_n^{*(b)} = T_n(X_1^{*(b)}, \dots, X_n^{*(b)})$ for each $b = 1, \dots, B$.
3. Approximate the sampling distribution of T_n by the empirical distribution of $\{T_n^{*(1)}, \dots, T_n^{*(B)}\}$.

A **bootstrap confidence interval** at level $1 - \delta$ uses the $\delta/2$ and $1 - \delta/2$ quantiles of $\{T_n^{*(b)}\}$.

4.2 Validity: asymptotic justification

Theorem 4.2 — Bootstrap Consistency (informal)

Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} L$ for some limit distribution L . Under regularity conditions (Hadamard differentiability of T), the bootstrap distribution $\sqrt{n}(T_n^* - T_n)$ also converges to L in probability:

$$\sup_{t \in \mathbb{R}} |\mathbb{P}^*(\sqrt{n}(T_n^* - T_n) \leq t) - \mathbb{P}(\sqrt{n}(T_n - \theta) \leq t)| \xrightarrow{p} 0,$$

where $\mathbb{P}^*$ denotes probability conditional on the data.

Remark — Bootstrap as a Bridge Between Paradigms

The bootstrap is asymptotically justified (Theorem 4.2) but produces *finite-sample* confidence intervals — a blend of the two paradigms.

Asymptotic features: Consistency holds as $n \rightarrow \infty$; the bootstrap exploits the limit distribution L .

Non-asymptotic features: At a fixed n the bootstrap returns an actual interval, not just a limiting statement, and its coverage probability is a definite (if unknown) number — generally not the nominal $1 - \delta$. The appeal is how fast that gap closes: for a *studentized* statistic, coverage error is $O(n^{-1})$ versus $O(n^{-1/2})$ for the normal approximation. Studentization, not smoothness, is what buys this — the plain percentile bootstrap is $O(n^{-1/2})$, like the CLT.

Key limitation: Bootstrap consistency can fail for non-smooth statistics (e.g., the sample maximum, the median in certain regimes) and in high dimensions. In these settings, non-asymptotic bounds (Hoeffding, Bernstein, Rademacher complexity) are more reliable.

Example 4.3 — Bootstrap vs. Delta Method for the Log Mean

Let $X_i \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$ and consider $T_n = \log \bar{X}_n$.

- **Delta method:** $g(x) = \log x$, $g'(x) = 1/x$. Asymptotic variance of T_n is $(g'(\mu))^2 \text{Var}(\bar{X}_n) = 1/(n\lambda^2\mu^2) = 1/n$. 95% CI: $\log \bar{X}_n \pm 1.96/\sqrt{n}$.
- **Bootstrap:** Draw $B = 1000$ bootstrap resamples, compute $\log \bar{X}_n^*$ each time, take empirical quantiles. For small n (say $n = 20$), the bootstrap CI is more accurate because the skewness of $\log \bar{X}_n$ is captured directly.

Both methods converge to the same CI as $n \rightarrow \infty$, but the bootstrap is more accurate for small n at the cost of computation.

5 Synthesis: A Decision Framework

5.1 When the two paradigms give the same answer

Remark 5.1 — Agreement in the Gaussian Regime

For i.i.d. sub-Gaussian variables with parameter σ^2 , both approaches give an $O(\sigma/\sqrt{n})$ rate for $\bar{X}_n$:

- **Non-asymptotic (Hoeffding/sub-Gaussian):** $|\bar{X}_n - \mu| \leq \sigma \sqrt{2 \log(2/\delta)/n}$ w.p. $\geq 1 - \delta$.
- **Asymptotic (CLT):** $|\bar{X}_n - \mu| \leq z_{\delta/2} \sigma / \sqrt{n}$ approximately w.p. $\geq 1 - \delta$.

For $\delta = 0.05$: $\sqrt{2 \log(40)} \approx 3.69$ vs. $z_{0.025} = 1.96$. The non-asymptotic bound is wider by a factor of ≈ 1.88 . This is the *price of distribution-freeness* — no approximation error, but a constant-factor loss.

Remark 5.2 — When the Two Paradigms Diverge

The paradigms give qualitatively different answers in three regimes:

1. **Heavy tails.** Non-asymptotic: use Bernstein with a truncation argument or Catoni's robust mean estimator. Rate: $O(1/\sqrt{n})$ still, but constant depends on $\mathbb{E}[|X|^{1+\alpha}]$ for some $\alpha > 0$. Asymptotic: CLT fails; require $\text{Var}(X) < \infty$; CLT rate is $O(1/\sqrt{n})$ but Gaussian approximation is inaccurate.
2. **High dimensions** ($d \asymp n$). Non-asymptotic: VC bound / Rademacher complexity give explicit rates in terms of d/n . Regularized methods (Lasso, ridge) have well-understood non-asymptotic guarantees. Asymptotic: MLE is inconsistent; χ^2 tests are anticonservative.
3. **Small n , rare events.** Non-asymptotic: Bernstein/Poisson bounds are valid and useful. Asymptotic: CLT is inaccurate; even Gaussian mixture approximations require $n \gg 1/p$ where p is the event probability.

5.2 The full comparison table

	Asymptotic paradigm	Non-asymptotic paradigm
Core question	What happens as $n \rightarrow \infty$?	What holds for fixed n ?
Primary tool	CLT, MLE theory, Delta method	Concentration inequalities
Output	Approximate CI, p -value, posterior	Explicit probability bound
Distributional assumption	Often parametric; $\text{Var} < \infty$	Often distribution-free
Validity	Approximate; improves with n	Exact; holds for all n
Sharpness	Sharp constants (e.g., $1.96\sigma/\sqrt{n}$)	Conservative constant (e.g., $3.69\sigma/\sqrt{n}$)
Failure modes	Misspecification; heavy tails; high d	Bounds loose; independence required
Application	Classical inference, econometrics	ML theory, high-dim., online learning
Middle ground	Bootstrap ($O(n^{-1})$ error if studentized)	—

5.3 A practical decision framework

Framework: Which Paradigm to Use?

Ask the following questions in order:

1. Is n large relative to the problem?

- If yes *and* the model is correctly specified $\Rightarrow$ **use asymptotic** (MLE, CLT-based CI).
- If no or uncertain $\Rightarrow$ continue.

2. Is the distribution heavy-tailed or completely unknown?

- If yes $\Rightarrow$ **use non-asymptotic** (Bernstein or sub-Gaussian bound with estimated σ).

3. Is the dimension d comparable to n ?

- If yes $\Rightarrow$ **use non-asymptotic** (VC / Rademacher bounds, regularized estimators with explicit rates).
4. **Is the statistic complex (quantile, ratio, functional)?**
- If yes and variance is hard to compute $\Rightarrow$ **use bootstrap** (asymptotically valid, finite-sample accurate; $O(n^{-1})$ error for smooth statistics, provided you studentize).
5. **Do you need a guaranteed coverage interval?**
- If yes (e.g., safety-critical systems) $\Rightarrow$ **use non-asymptotic** regardless of n .

5.4 Are the two paradigms competitors or complements?

Remark 5.3 — Complementarity, Not Competition

The two paradigms answer different questions:

- **Asymptotic theory** tells us the *shape* of the sampling distribution as n grows, yielding sharp, interpretable inference tools (CIs, hypothesis tests, model selection criteria). It is the right language for understanding *efficiency*.
- **Non-asymptotic theory** tells us *how much data is sufficient* for a given guarantee, without any reference to a limit. It is the right language for understanding *learning and generalization*.
- In modern statistics, both are used simultaneously. A statistician might use an asymptotic test for interpretability while verifying the result against a Hoeffding bound as a sanity check. A machine learning theorist might prove a VC generalization bound and then use the CLT to understand its tightness as $n \rightarrow \infty$.

The deepest results in modern statistics (e.g., adaptive estimation, minimax theory, information-theoretic lower bounds) require *both* paradigms working together.

6 Summary of the Lecture Series

What we proved

L1	WLLN, CLT	Asymptotic foundations; Markov, Chebyshev as first non-asymptotic tools.
L2	MLE theory	Consistency (KL/WLLN), asymptotic normality (score + CLT + Slutsky), Delta method, three failure modes.
L3	Concentration inequalities	Cramér–Chernoff method, Hoeffding’s lemma and inequality, Bernstein’s inequality, sub-Gaussian/sub-exponential families.
L4	Union bound + max bound + synthesis	Union bound, Bonferroni correction, $\sqrt{\log p}$ multiplicity cost, maximum of sub-Gaussians ($\mathbb{E}[\max X_j] \leq \sigma\sqrt{2\log p}$), Gumbel limit, bootstrap, decision framework.

The central message

Central Message of the Course

The asymptotic and non-asymptotic paradigms are two languages for quantifying statistical uncertainty:

- Asymptotic: sharp, model-based, interpretable — relies on $n \rightarrow \infty$.
- Non-asymptotic: exact, distribution-free, dimension-aware — valid for all n , at the cost of larger constants.

A complete statistician speaks both languages fluently, uses each where it is strongest, and understands precisely where and why they disagree.

7 Exercises

Exercise 7.1 (Union bound: simultaneous testing). Let $X_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ for $j = 1, \dots, p$ and $i = 1, \dots, n$, so all null hypotheses $H_j : \mu_j = 0$ are true.

- Show that the probability of at least one false rejection at individual level α satisfies $\mathbb{P}(\text{any rejection}) \leq p\alpha$. For $p = 100$ and $\alpha = 0.05$, what is the bound?
- The **Bonferroni correction** uses individual level α/p . Show this controls the familywise error rate (FWER) at level α .
- Show that the Bonferroni threshold $z_{\alpha/(2p)}$ satisfies $z_{\alpha/(2p)} \approx \sqrt{2\log(p/\alpha)}$ for large p . (Hint: use the Gaussian tail bound $1 - \Phi(t) \leq e^{-t^2/2}$.)

- (d) For $p = 100$, $\alpha = 0.05$: compute $z_{\alpha/2}$ (standard) vs. $z_{\alpha/(2p)}$ (Bonferroni). What is the ratio, and how does it scale with p ?

Exercise 7.2 (Maximum of sub-Gaussian variables). Let $X_1, \dots, X_p \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2)$.

- (a) Prove that $\mathbb{E}[\max_j X_j] \leq \sigma\sqrt{2\log p}$ using the Laplace transform trick from Theorem 3.2, and verify the optimal $t^* = \sqrt{2\log p}/\sigma$.
- (b) Show that the bound is tight: for Gaussians, $\mathbb{E}[\max_j X_j]/(\sigma\sqrt{2\log p}) \rightarrow 1$ as $p \rightarrow \infty$. (Hint: $\mathbb{E}[\max_j X_j] \geq u \cdot \mathbb{P}(\max_j X_j \geq u)$ for any u ; choose $u = \sigma\sqrt{2\log p - \varepsilon}$.)
- (c) Let $M_p = \max_{1 \leq j \leq p} X_j$ and $u = \sigma\sqrt{2\log(p/\delta)}$. Using Theorem 3.1, find the smallest p such that $\mathbb{P}(M_p \geq u) \leq 0.05$ when $\sigma = 1$ and $\delta = 0.05$.

Exercise 7.3 (Bootstrap coverage). Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exponential}(\lambda)$ with $n = 30$.

- (a) Construct a 95% CI for λ using the Delta method applied to $T_n = 1/\bar{X}_n$ (the MLE of λ).
- (b) Describe the bootstrap procedure for constructing a 95% CI for λ .
- (c) For $\lambda = 1$: using the exact distribution of T_n , what is the true coverage of the Delta method CI? (Remark: The bootstrap corrects for skewness that the Delta method ignores.)

Exercise 7.4 (The $\sqrt{\log p}$ factor: necessity). This exercise shows the $\sqrt{\log p}$ factor in Theorem 3.1 cannot be removed, even with the best possible bound. Let $X_1, \dots, X_p \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$.

- (a) Using the union bound and the exact Gaussian tail $\mathbb{P}(X \geq t) \leq \frac{1}{t\sqrt{2\pi}}e^{-t^2/2}$, show: $\mathbb{P}(\max_j X_j \geq t) \leq \frac{p}{t\sqrt{2\pi}}e^{-t^2/2}$.
- (b) Set the right-hand side equal to δ and solve for t . Confirm you recover $t \approx \sqrt{2\log(p/\delta)}$.
- (c) For $p = 10, 100, 1000, 10^6$: compute the 95% threshold from the union bound and compare with the exact simulation value. How good is the approximation?

Exercise 7.5 (Asymptotic vs. non-asymptotic rates: a case study). Suppose $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with μ unknown, σ^2 known.

- (a) Write down the exact $1 - \delta$ confidence interval for μ . (For Gaussian data with known σ^2 , this is exact for all n .)
- (b) Write down the non-asymptotic CI using the sub-Gaussian bound.
- (c) For $n = 10$, $\delta = 0.05$: compute the width of each CI. What is the ratio of non-asymptotic to exact width?
- (d) Show that as $\delta \rightarrow 0$ (very high confidence), the ratio of the two CI widths converges to 1. Interpret this.

(e) For $n = 10,000$: how do the two CIs compare? When is the sub-Gaussian bound competitive?

Exercise 7.6 (Putting it all together: the $\sqrt{\log p}$ factor in practice). A genomics study measures the expression levels of $p = 20,000$ genes. For each gene j , we observe $n = 50$ samples $X_{ij} \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu_j, 1)$ and test $H_j : \mu_j = 0$ against $H'_j : \mu_j \neq 0$.

- (a) Using Bonferroni, what individual-level threshold controls FWER at 0.05?
- (b) What is the corresponding sub-Gaussian bound on $\mathbb{P}(\max_j |\bar{X}_j| \geq t)$ for the threshold t from (a)?
- (c) If instead only $\mu_j = 0$ for all j (global null), what is the exact probability that the largest $|\bar{X}_j|$ exceeds the Bonferroni threshold? Is Bonferroni conservative here?
- (d) A colleague suggests using individual level $\alpha = 0.05$ with no correction. Compute the expected number of false rejections. Contrast with the Bonferroni approach.

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